

Degradable Vinyl Polymers for Combating Marine Biofouling

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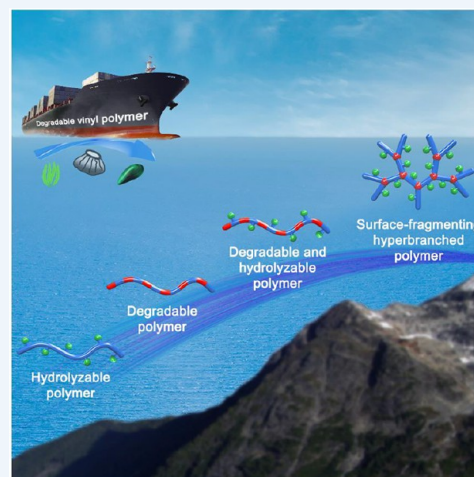
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CONSPECTUS: Marine organisms such as barnacle larvae and spores of algae adhere to underwater surfaces leading to marine biofouling. This phenomenon has numerous adverse impacts on marine industries and maritime activities. Due to the diversity of fouling organisms and the complexity of the marine environment, it is a huge challenge to combat marine biofouling, which limits the development and utilization of marine resources. Since the International Marine Organization banned the use of tributyltin self-polishing copolymer (SPC) coatings in 2008, the development of an environmentally friendly and efficient anti-biofouling polymer has been the most important task in this field. Tin-free SPC is a well-established and widely used polymer binder for anti-biofouling coating today. Being a nondegradable vinyl polymer, SPC exhibits poor anti-biofouling performance in static conditions. Even more, such nondegradable polymers were considered to be a source of microplastics by the International Union for the Conservation of Nature in 2019. Recently, numerous degradable polymers, which can form dynamic surface through main chain scission, have been developed for preventing marine biofouling in static conditions. Nevertheless, the regulation of their degradation and mechanical properties is limited, and they are also difficult to functionalize. A new polymer combining the advantages of vinyl polymers and degradable polymers is needed. However, such a combination is a challenge since the former are synthesized via free radical polymerization whereas the latter are synthesized via ring-opening polymerization.

In this Account, we review our recent progress toward degradable vinyl polymers for marine anti-biofouling in terms of polymerization methods and structures and properties of polymers. First, we introduce the strategies for preparing degradable vinyl polymers with an emphasis on hybrid copolymerization. Then, we present the synthesis and performance of degradable and hydrolyzable polyacrylates, degradable polyurethanes with hydrolyzable side groups, and surface-fragmenting hyperbranched polymers. Polymers with degradable main chains and hydrolyzable side groups combine the advantages of SPC and degradable polymers, so they are degradable and functional. They are becoming new-generation polymers with great potential for preparing high-efficiency, long-lasting, environmentally friendly and broad-spectrum coatings to inhibit marine biofouling. They can also find applications in wastewater treatment, biomedical materials, and other fields.



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An Eco-Friendly Antifouling Coating Consisting of Poly(lactic acid)-Based Polyurethane and Natural Antifoulant. *ACS Sustainable Chem. Eng.* **2020**, 8, 1671–1678.³ This work proposed an instructive strategy for anti-biofouling "from the nature, for the nature" and reported a degradable and hydrolyzable polyurethane based ecofriendly coating.

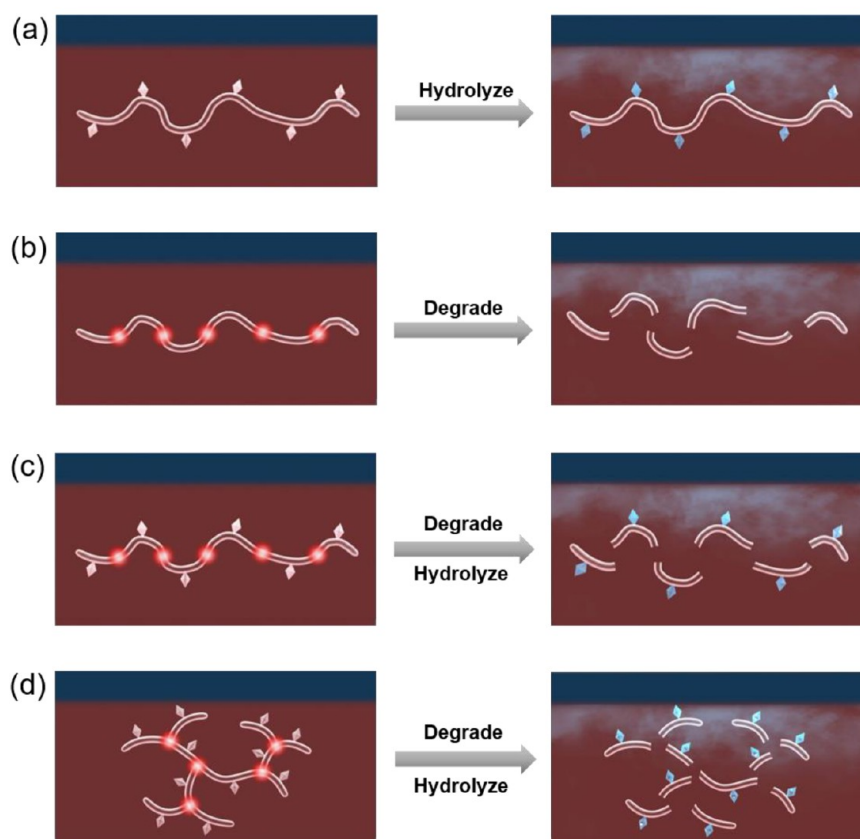
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Scheme 1. Anti-biofouling Mechanisms of (a) Hydrolyzable Polymer, (b) Degradable Polymer, (c) Degradable and Hydrolyzable Polymer, and (d) Surface-Fragmenting Hyperbranched Polymer



Self-Regenerating Attack and Defense for Antifouling Coatings. *ACS Appl. Mater. Interfaces* **2021**, 13, 13735–13743.⁴ This work reported a degradable hyperbranched polymer with self-generating antifoulant and zwitterionic groups, which integrates fouling organism killing, resistance, and releasing functions to combat marine biofouling.

1. INTRODUCTION

Marine biofouling is caused by the persistent adhesion and reproduction of marine microorganisms, plants, and animals on the surface of an entity submerged in seawater. It has numerous adverse impacts on maritime transportation as well as the development and utilization of marine resources.⁵ For example, marine biofouling increases a ship's sailing resistance and hence decreases its sailing speed. It also increases fuel consumption and carbon dioxide emissions. Marine biofouling can change the structure and state of the ship hull surface inducing and accelerating its corrosion.⁶ Macroscopic marine animals and plants can block the mesh and thus reduce the exchange of oxygen and nutrients of mariculture cages. As a result, the biofouling reduces the production of a sea ranch and may even result in massive death of fish and shrimp there.⁷ According to an incomplete survey, the global economic cost of marine biofouling is tens of billions of dollars a year. Biofouling has attracted increasing attention as the maritime industry develops and human activity in the oceans increases.

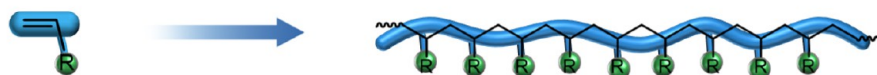
To combat marine biofouling, mechanical cleaning, electrochemistry, and ultrasonic techniques have been used. Yet, the most convenient, economical, and widely used approach is anti-

biofouling coatings. Since the International Marine Organization banned the use of tributyltin self-polishing copolymer (SPC) coatings in 2008, the development of environmentally friendly and efficient marine anti-biofouling coatings has been the most important task in this field. Accordingly, tin-free SPC coatings and fouling release coatings,^{8,9} as well as degradable polymers,¹⁰ amphiphilic polymers,¹¹ biomimetic polymers,¹² and protein-resistant polymers¹³ have been developed for this purpose. SPC consists of a vinyl polymer with hydrolyzable side groups such as silyl, copper, and zinc ester that can react with seawater in a controlled manner (Scheme 1a). Their unstable side groups hydrolyze under the attack of seawater so that the polymers become hydrophilic and are then polished away by strong water flow. The anti-biofouling performance increases with vessel rate, and this polymer exhibits poor performance in static conditions. Moreover, since the SPC main chain does not degrade, once the polymer fragments enter the ocean, they form so-called marine microplastics as the International Union for the Conservation of Nature stated in 2019.¹⁴

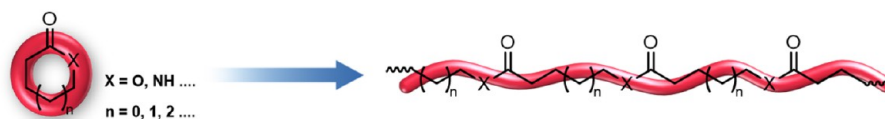
Recently, degradable polymers have been reported to lessen biofouling adhesion because they form a dynamic surface or continuously changing surface, which can significantly reduce the contact time and force between the organisms and the surfaces.^{15,16} The polymers can continuously renew their surfaces in a marine environment via the cleavage of main chains even under static conditions (Scheme 1b). Such cleavage only takes place on the surface in contact with seawater and enzymes where the polymer degrades into small molecules that are easily dispersed in seawater, thereby they do not form marine

Scheme 2. Schematic of (a) Radical Polymerization of Vinyl Monomers, (b) Ring-Opening Polymerization of Cyclic Monomers, and (c) Hybrid Copolymerization with Various Typical Cyclic and Vinyl Monomers

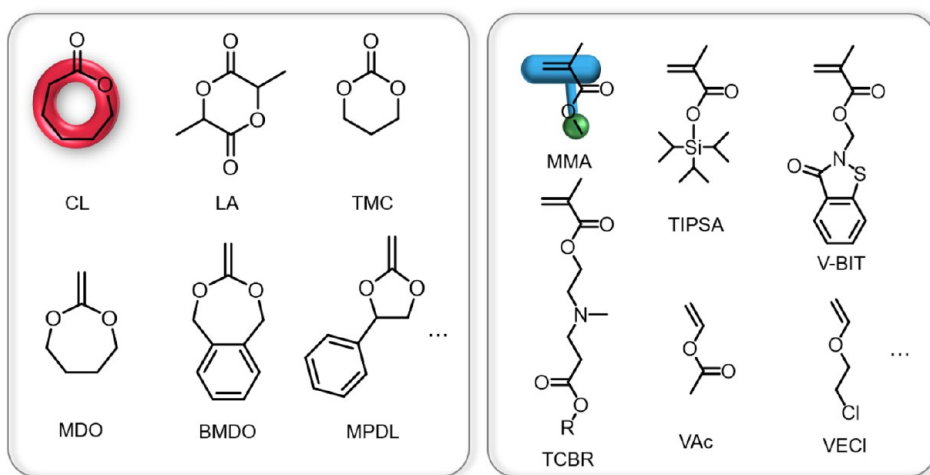
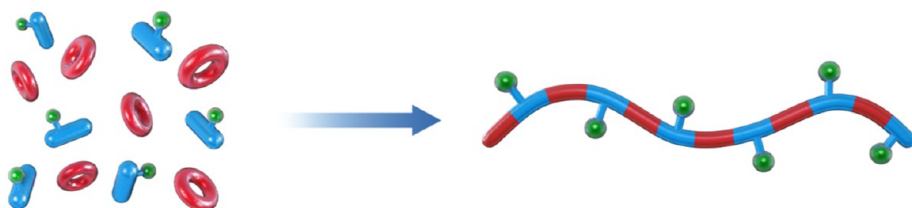
(a) Radical polymerization of vinyl monomers



(b) Ring-opening polymerization of cyclic monomers



(c) Hybrid copolymerization



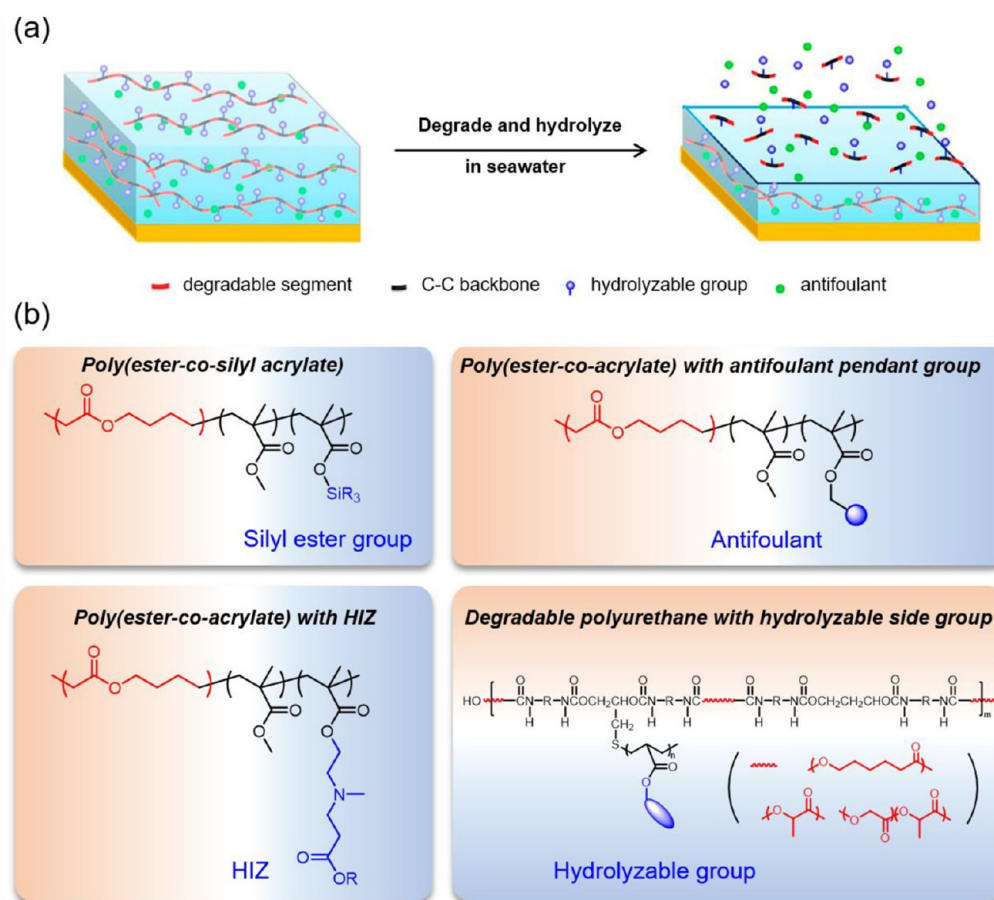
microplastics. Generally, the anti-biofouling ability of degradable polymers increases with the density of ester and hydrophilicity but decreases with crystallinity, which is contrary to the response of the mechanical properties. Thus, the window for regulating degradation and mechanical properties of degradable polymers is narrow. On the other hand, degradable polymers are difficult to functionalize or chemically modify. This is because a cyclic monomer generally cannot have functional or modifiable groups to stabilize the cyclic structure. Accordingly, the number of functional cyclic monomers is limited, and it is difficult to synthesize cyclic monomers with modifiable groups. A combination of vinyl polymer and degradable polymer is expected to exhibit a wide spectrum of properties. Generally, vinyl polymer is prepared via free radical polymerization of vinyl monomers (Scheme 2a), whereas degradable polymer is synthesized via ring-opening polymerization of cyclic monomers (Scheme 2b). It is a challenge to copolymerize or chemically connect vinyl and cyclic monomers into one polymer chain because of their quite different polymerization mechanisms and reaction activities.

We have synthesized various degradable vinyl polymers via hybrid copolymerization, reversible addition–fragmentation chain transfer (RAFT) polymerization, or a combination of thiol–ene reaction or telomerization and polyaddition. The functional side groups from vinyl monomers and the polyester segments from cyclic monomers can readily be connected into the same polymer chain via the polymerizations. So, the polymer exhibits a combination of properties of hydrolyzable vinyl polymers and degradable non-vinyl polymers. Moreover, these polymers can be linear or hyperbranched (Scheme 1c,d). They have a cleavable main chain and hydrolyzable side groups such as silyl esters, antifoulants, and hydrolysis-induced zwitterions (HIZs). The cleavable bond makes it easy for the polymer to be attacked by seawater or enzymes in static conditions and degrade into small molecules, which readily dissolve in seawater, forming a dynamic surface. Meanwhile, the hydrolyzable side groups transform from hydrophobic to hydrophilic after hydrolysis, increasing the solubility of the polymer and thereby improving the self-polishing rate and the anti-biofouling ability of the coating. With those coordinating mechanisms, dynamic anti-biofouling surfaces are more sustainable and controllable.

Table 1. Mechanisms, Advantages, Disadvantages, and Challenges of Different Types of Anti-biofouling Polymers

polymer type	mechanisms	advantages	disadvantages, challenges
hydrolyzable polymer	hydrolysis of side groups (silyl, copper, zinc ester)	good anti-biofouling performance under strong water flow	poor anti-biofouling performance in static conditions; forming marine microplastics
degradable polymer	cleavage of main chain ester	good anti-biofouling performance in static conditions; does not form marine microplastics	hard to balance degradation and mechanical properties; difficult to functionalize or chemically modify
degradable and hydrolyzable polymer	degradation of main chains and hydrolysis of side groups	good anti-biofouling performance in static conditions; multiple functions, for example, intelligent response, drag reduction, and anticorrosion	need to match the vinyl monomer and cyclic monomer
surface-fragmenting hyperbranched polymer	degradation of main chains, cleavage of branching units, or both	good anti-biofouling performance in static conditions; lower molecular weight of degradation products; high solid content and low viscosity	need to well control the branching degree

Scheme 3. (a) Anti-biofouling Mechanism of Degradable and Hydrolyzable Polymers and (b) Structures of Degradable and Hydrolyzable Polymers



Panel a adapted with permission from ref 29. Copyright (2015) American Chemical Society.

Furthermore, green antifoulants can be combined into the polymers to further strengthen the anti-biofouling ability and prolong the service life of the coating. In comparison with traditional SPCs, the degradable vinyl polymer surface shows little dependence of renewing rate on water flow. Even in static conditions, they are able to degrade completely into small molecules and form a controllable dynamic surface in the presence of seawater, so they have high surface self-renewal rate and excellent anti-biofouling performance. In comparison to degradable polymers, degradable vinyl polymers have more controlled degradation and improved mechanical properties, and they can be more readily functionalized. [Table 1](#) summarizes the mechanisms, advantages, disadvantages, and challenges of different types of anti-biofouling polymers. In this Account, we

present the design and preparation of degradable vinyl polymers in terms of synthesis, structures, and properties of the polymers. We also provide a perspective on potential future developments in this field.

2. SYNTHESIS METHODS FOR DEGRADABLE VINYL POLYMERS

Degradable vinyl polymers have been synthesized via controlled radical polymerization including nitroxide-mediated polymerization (NMP), atom-transfer radical polymerization (ATRP), and RAFT polymerization.¹⁷ A difunctional initiator with cleavable function is a prerequisite in these polymerizations, so that the obtained polymer has a single cleavable unit,¹⁸ degradable blocks,¹⁹ or cleavable side chains.²⁰ The procedures

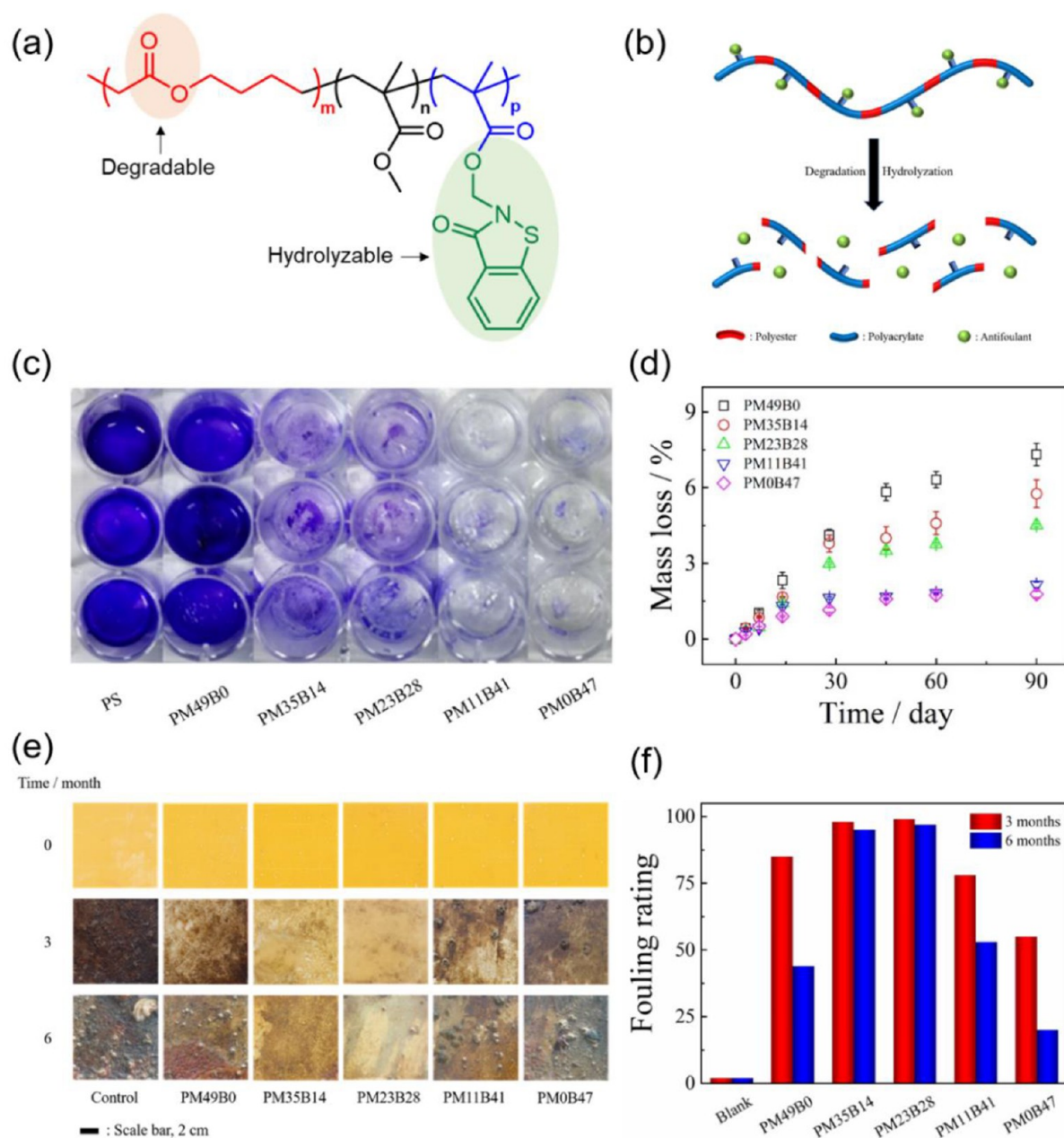


Figure 1. (a) Structure of poly(ester-co-acrylate) with BIT pendant groups. (b) Degradation and hydrolyzation mechanism of the polymers. (c) Biofilm of *Pseudomonas* sp. grown on the coating surfaces stained with crystal violet. (d) Variation of mass loss of the polymers in ASW at 25 °C. (e) Images and (f) fouling rating of the coatings after immersion in Xiamen Bay for 6 months. Adapted with permission from ref 31. Copyright 2019 American Chemical Society.

are usually tedious and complicated, and the resultant polymers are only partly degradable. To incorporate more dispersed cleavable segments into the main chains, several new polymerization methods have been developed including hybrid copolymerization of cyclic monomers (cyclic ketene acetals and lactone) with vinyl monomers, controlled copolymerization of cleavable divinyl monomers, and the combination of thiol–ene reaction or telomerization and polyaddition.

Hybrid copolymerization refers to a reaction where two or more structurally dissimilar monomers (e.g., cyclic monomers and vinyl monomers) copolymerize into the same chain (Scheme 2c).²¹ It can produce new types of degradable vinyl polymers. Hybrid copolymerization has been known for a long

time, but only a few breakthroughs have been made so far. The radical ring-opening hybrid copolymerization of 2-methylene-1,3-dioxepane (MDO) with vinyl monomers was first reported by Bailey in 1980s, where MDO can form polycaprolactone (PCL) segments via ring-opening.²² The above NMP, ATRP, and RAFT polymerizations have been used in the radical ring-opening hybrid copolymerization.²³ However, the cyclic monomers for the polymerization are limited to cyclic ketene acetals (CKAs). Copolymerization of CKAs with classic vinyl monomers leads to the incorporation of cleavable ester bonds into a vinyl polymer backbone. We have prepared a degradable copolymer of MDO and methyl methacrylate (MMA) for marine anti-biofouling via radical ring-opening hybrid copoly-

merization, whose degradation rate can be regulated by the content of the ester segment.²⁴ Our group first reported the anionic hybrid copolymerization of ϵ -caprolactone (CL) and MMA in 2012,¹ where phosphazene (*t*-BuP₄) was used as the catalyst and the resultant copolymer is random. Actually, many pairs of vinyl monomers and cyclic monomers can copolymerize via anionic hybrid copolymerization, for example, 2-(*N,N*-dimethylamino)ethyl methacrylate with CL and 2-(2-methoxyethoxy)ethyl methacrylate with L-lactide.^{25–27} The polymers exhibit controllable degradation rates and excellent protein resistance. Clearly, hybrid copolymerization has the capacity to prepare functional polyesters with active side groups and degradable main chains.

Furthermore, controlled copolymerization of cleavable divinyl monomers is a facile and versatile approach to synthesize degradable vinyl polymers with hyperbranched structures, which tolerates a wide range of monomers and various functionalities but can happen in relatively mild reaction conditions. Usually, RAFT polymerization and reversible complexation mediated polymerization (RCMP) are used to prepare hyperbranched polymers consisting of vinyl monomers, di- or multivinyl monomers, and chain transfer agents. Degradable vinyl polymers can also be prepared by combining thiol–ene reaction or telomerization and polyaddition, where the vinyl precursor with a dihydroxy at one end and different molecular weights is prepared and used as a chain extender to synthesize polyester-based polyurethane. These approaches provide strategies for synthesis of degradable vinyl polymers.

3. DEGRADABLE AND HYDROLYZABLE POLYMERS

In this part, we summarize the synthesis of a series of degradable vinyl polymers with degradable main chains and hydrolyzable side groups for combating marine biofouling (Scheme 3). We attempt to incorporate hydrolyzable silyl ester groups, antifoulants, hydrolysis-induced zwitterions, as well as other special functionalities, into the polymers.

3.1. Poly(ester-co-silyl acrylate)

Hybrid copolymerization is a facile approach to realize the incorporation of degradable ester main chains onto poly(silyl acrylate).²⁸ We first reported poly(ester-co-silyl acrylate) synthesized via radical ring-opening hybrid copolymerization of MMA, tributyl silyl methacrylate (TBSM), and MDO, and the polymer has already been used in a marine anti-biofouling coating.²⁹ The polymer backbone is degradable, and the side groups are hydrolyzable. The degradation and hydrolysis rates can be regulated by optimizing the molar ratio of ester units on the backbone and the tributyl silyl ester side groups, as well as the structure of the leaving group. We used MDO to react with three different silyl acrylates (bis(trimethylsiloxy)methylsilyl methacrylate (MATM2), triisopropylsilyl methacrylate (TIPSM), and TBSM) to prepare degradable copolymers with different hydrolysis rates (MATM2 > TBSM > TIPSM).³⁰ After immersion in artificial seawater (ASW) for 7 weeks, the copolymer containing MATM2 exhibits a noteworthy reduction in molecular weight of 90%, whereas the copolymer with TIPSM side groups did not show a notable loss in molecular weight. Clearly, rapid hydrolysis of silyl ester groups leads to quick generation of the hydrophilic carboxyl groups, so that the surface has high water absorption, which further accelerates the breakdown of the main chain. Accordingly, by optimizing the polymer molecular structure, we can obtain a renewable polymer with controllable degradability and hydrolyzability. Attacked by

enzymes and seawater, the polymers degrade into small molecules in a marine environment, so they do not become marine microplastics.

3.2. Poly(ester-co-acrylate) with Antifoulant Pendant Groups

In order to improve anti-biofouling performance, much effort has been made to synthesize degradable polyacrylates with new functions. Degradable poly(ester-co-silyl acrylate) has limited anti-biofouling performance since it does not contain any anti-biofouling groups. Antifoulants are usually added to improve the anti-biofouling performance in practical applications. Actually, the anti-biofouling functional groups can be readily introduced into the polymer via hybrid copolymerization. We synthesized an *N*-methacryloyloxymethyl benzisothiazolinone (BIT) monomer and prepared a poly(ester-co-acrylate) with antifoulant pendant groups via hybrid copolymerization of BIT, MDO, and MMA (Figure 1a,b).³¹ The copolymer with 47 wt % ester units loses 7.3 wt % mass after immersion in ASW for 90 days, while the copolymer without ester units shows only a 1.6 wt % mass loss (Figure 1d). The copolymer with 47 wt % BIT shows excellent antiadhesion performance toward bacteria and diatoms in laboratory bioassays (Figure 1c). Marine field tests show that the polymer with 23 wt % ester units and 28 wt % BIT exhibits the best fouling rating of 95, indicating that only 5% of the area of the coating surface is covered by macrofouling according to ASTM D6990-05 (2011) (Figure 1e,f). This is because the ester content affects the surface self-renewal and the release of BIT contributes to the fouling resistance; both of these properties are critical for anti-biofouling efficacy. In other words, the anti-biofouling performance of the main-chain degradable and side-group hydrolyzable renewable polymers is the result of the degradation cooperating with the hydrolysis.

3.3. Poly(ester-co-acrylate) with Hydrolysis-Induced Zwitterions

It is well-known that zwitterionic polymers are environmentally friendly and can form a stable hydration layer, which results in excellent anti-biofouling capacity.³² However, their high hydrophilicity makes the coating swell considerably in seawater, leading to a decrease in its mechanical properties. Also, their higher polarity makes them difficult to copolymerize with nonpolar monomers, which greatly limits their applications in marine anti-biofouling. On the other hand, the aforementioned zwitterionic polymers are not degradable. Once their surfaces are covered by inorganic substances and dead bacteria in a marine environment, they would lose their anti-biofouling function. Degradable polymers with hydrolysis-induced zwitterions are expected to solve the problem.

We have developed a class of zwitterionic precursor monomers called tertiary carboxybetaine (TCB) esters. They are hydrophobic but become hydrophilic zwitterions after hydrolysis.² The radical ring-opening hybrid copolymerization of TCB and MDO was used to prepare poly(ester-co-acrylate) with HIZs. Such a polymer can undergo cleavage of the ester bonds under the attack of seawater to form a constantly renewing surface, and its side groups hydrolyze to generate zwitterions, which allows the surface to have antiprotein, antibacterial, and antidiatom properties. To further regulate the formation rate of zwitterions, we synthesized a novel HIZ monomer with a faster hydrolysis rate, tertiary carboxybetaine triisopropylsilyl ester ethyl acrylate (TCBSA), and copolymerized it with MDO and MMA to prepare a self-generating and self-renewing zwitterionic polymer (Figure 2a).³³ The copoly-

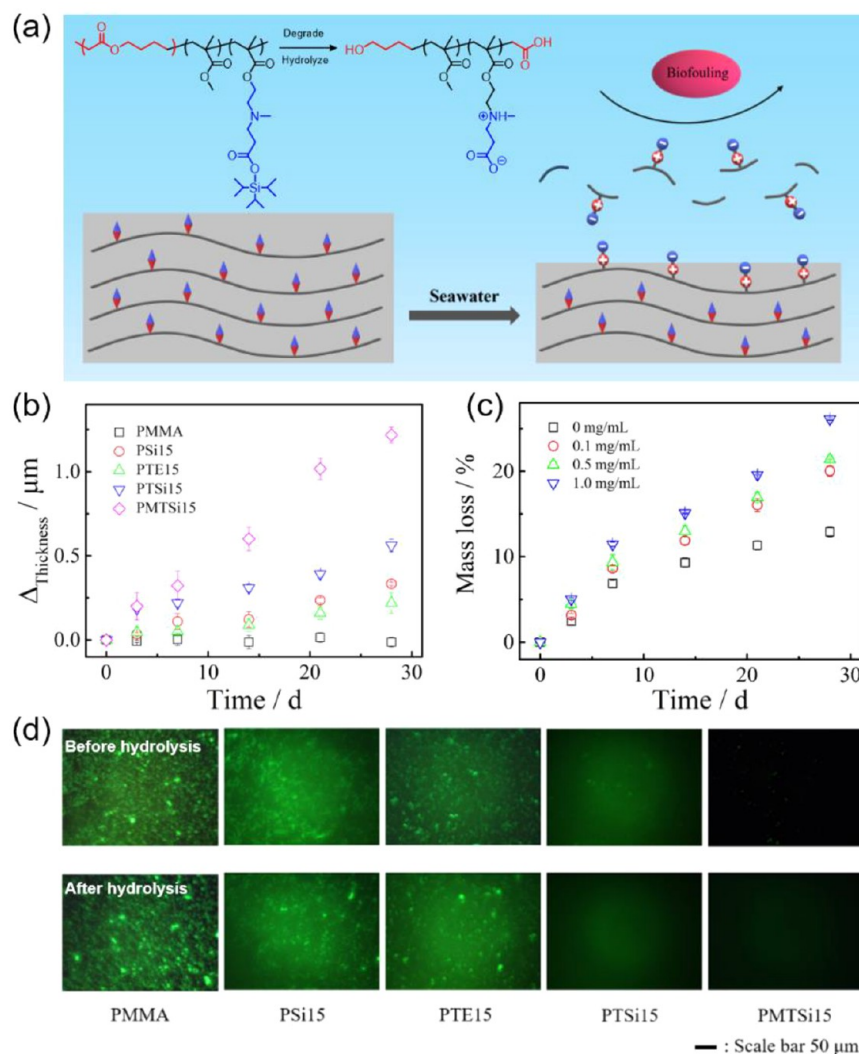


Figure 2. (a) Structure and anti-biofouling mechanism of poly(ester-co-acrylate) with HIZ. (b) Thickness loss of polymeric coatings in ASW. (c) Mass loss of typical polymer under different concentrations of lipase in ASW. (d) Fluorescence microscopy photographs of *Pseudomonas* sp. adhered on copolymer surfaces without and with prehydrolysis for 3 days. Adapted with permission from ref 33. Copyright 2019 American Chemical Society.

mer rapidly generates a hydrophilic zwitterionic surface with a water contact angle of $\sim 67^\circ$ through the induction with seawater for 7 days, whereas the hardness grade still remains at H and HB. Namely, the hydrolysis only occurs on the coating surface, and the interior is still hydrophobic and holds its mechanical strength. Owing to the self-renewing zwitterionic surface, this polymeric coating can effectively inhibit the adhesion of protein, marine bacteria, and diatoms (Figure 2b,c,d). Clearly, self-generating and self-renewing zwitterionic polymers are promising in anti-biofouling coatings.

3.4. Degradable Polyurethanes with Hydrolyzable Side Groups

Polyurethane is widely used in coatings, adhesives, food packaging, building materials, and medical materials because of its excellent mechanical properties and good adhesion to a substrate.^{34,35} As regards marine anti-biofouling, polyurethane with high adhesion strength and capacity for large deformations is irreplaceable for flexible substrates such as rubber and fishing net.³⁶ In recent years, we have introduced degradable segments into polyurethane to prepare a series of degradable polyurethanes for marine anti-biofouling.^{37–41} Following this concept,

triblock and star-shaped polyester based polyurethanes were developed to inhibit marine biofouling.^{42,43} Those polyurethanes not only have adequate mechanical properties to meet the demands of flexible substrates but also degrade at a constant rate, so that the release of antifoulants in static or dynamic conditions is controlled.

We introduced biodegradable PCL into poly(triisopropylsilyl acrylate) (PTSA) to prepare a main-chain degradable and side-group hydrolyzable polyurethane.⁴⁴ The polymer forms a self-renewing dynamic surface in a marine environment via degradation and hydrolyzation, which can effectively inhibit the adhesion of fouling organisms even in static conditions. Degradable polymer also acts as a carrier of antifoulants, which makes the controlled release of the antifoulants possible, further improving the antifouling ability. In short, the antifouling performance becomes better as the degradation rate increases. Yet, the antifouling duration decreases with the degradation rate. To have good anti-biofouling performance with long service life, the antifoulant 4,5-dichloro-2-*n*-octyl-3-isothiazolinone (DCOIT) was added to prepare a coating. It is highly active on the coating surface and can rapidly degrade in marine environments. The coating showed excellent anti-biofouling

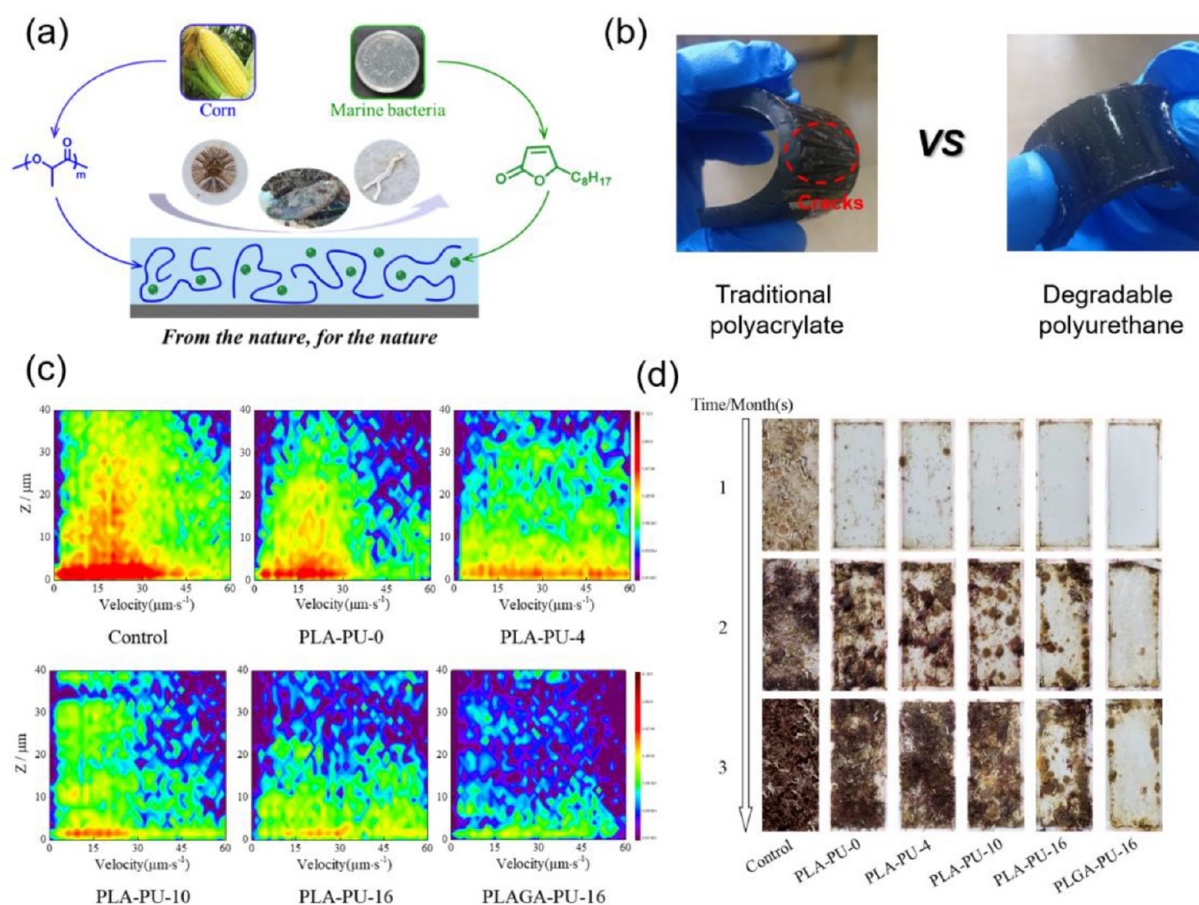


Figure 3. (a) Anti-biofouling mechanism of ecofriendly coating with PLA-based polyurethane and natural antifoulant. (b) Pictures of traditional polyacrylate and degradable polyurethane coated on rubber after bending. (c) Heat maps of instantaneous 3D velocities and distances from the coating surfaces for *Pseudomonas* sp. Color bars represent the relative density of data points with a logarithmic scale. (d) Images of coatings with 10 wt % butenolide after immersion in natural seawater for 3 months. Adapted with permission from ref 3. Copyright 2020 American Chemical Society.

performance in long-term static marine field tests in the South China Sea and the East China Sea. We also prepared polyurethanes containing different polyester (PCL, poly(L-lactide) and poly(ethylene adipate)) segments and PTSA side chains with different length.⁴⁵ The degradation rate of the main chain and the hydrolysis rate of the side chain can be regulated by altering the ester density and PTSA molecular weight, making control of the antifoulant release possible. Clearly, as the content of ester bonds and hydrolyzable side groups increases, the degradation rate increases and the static anti-biofouling performance is improved. The polymer combined with DCOIT showed excellent anti-biofouling performance in a marine environment because of the synergetic effect of the surface renewal and the release of antifoulant.

"From the nature, for the nature" is an effective strategy that has been used to develop environmentally friendly anti-biofouling coatings (Figure 3a).³ The raw materials used in the synthesis of poly(L-lactide) (PLA) predominantly come from crops. It is one of the most widely used biosource-based and biodegradable polyesters. PLA can quickly and completely degrade into nontoxic carbon dioxide and water and finally return to nature. Through a thiol–ene reaction and poly-addition, we prepared biosourced PLA-based polyurethane containing hydrolyzable triisopropylsilyl acrylate (TSA) side groups. This polymer is flexible with a high adhesion strength (Figure 3b), and its degradation rate can be adjusted by

changing its soft segment and TSA content. An environmentally friendly coating can be prepared by compounding the polyurethane with natural antifoulant (butenolide), which is obtained from marine bacteria and can be described as being "from the sea, for the sea". In such a system, the butenolide can be continuously released from the coating at a controllable and stable rate via continuous degradation of the polymer. It can decrease the irreversible adhesion of bacteria and realize highly effective antibacterial function but does not kill them (Figure 3c). As shown in Figure 3d, the coating exhibited excellent anti-biofouling performance. The polymer can be used to develop high-performance and green anti-biofouling coatings.

4. SURFACE-FRAGMENTING HYPERBRANCHED POLYMERS

Degradable hyperbranched polymers consisting of linear units, branching units, and terminal groups have three-dimensional polydisperse and ellipsoidal random branched structures. Compared with linear degradable polymers, their degradation occurs on either main chains or branching points, and the degradation products have lower molecular weight, so they more readily realize surface self-renewal than linear ones. Hyperbranched polymers also have many advantages over linear ones, including high solubility (permitting the preparation of coating with high solid content but low viscosity), multiple modifiable terminal groups, and good film-forming properties. Thus,

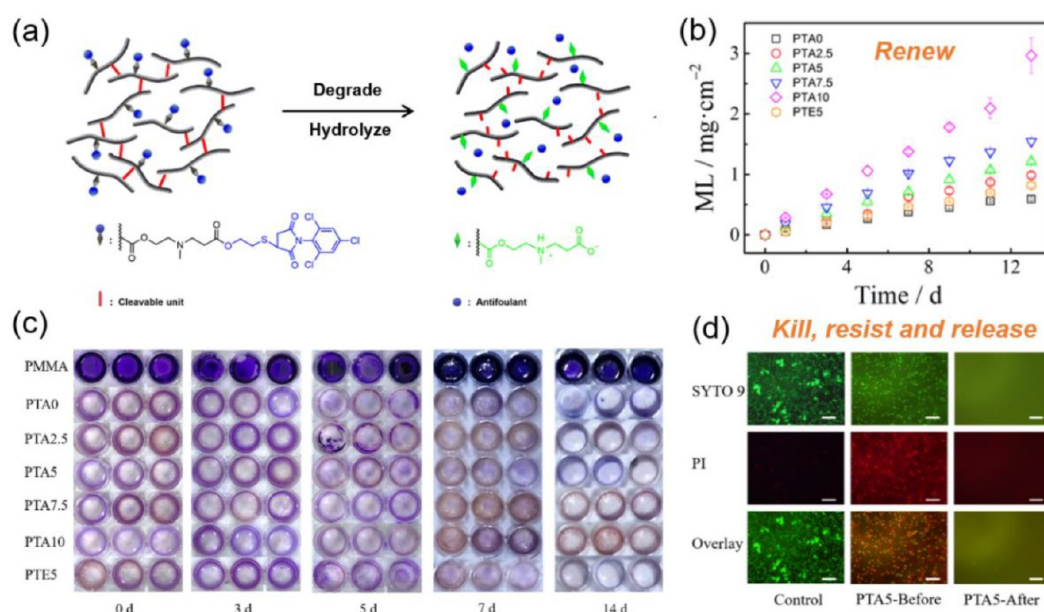


Figure 4. (a) Scheme for the degradation and hydrolysis process of hyperbranched polymers with cleavable MAAH as branching units. (b) Mass loss of the polymers in ASW at 25 °C. (c) Biofilm of *Pseudomonas* sp. grown on the polymeric coating surfaces after different prehydrolysis time. (d) Fluorescence microscopy photographs of *Pseudomonas* sp. adhered on the polymeric coating surfaces. Adapted with permission from ref 4. Copyright 2021 American Chemical Society.

hyperbranched polymers have a broad spectrum of application in anti-biofouling coatings.⁴⁶ In the following sections, we present degradable vinyl polymers from linear to hyperbranched, which can degrade into smaller hydrophilic fragments and more readily dissolve in seawater, yielding a self-renewing and surface-fragmenting anti-biofouling (SFA) polymer.

4.1. Hyperbranched Polymers with Cleavable Branching Units

A degradable hyperbranched polymer with divinyl-terminated polycaprolactone (PCL- vi_2) as a branch bridge between the main chain of PMMA was prepared by RCMP.⁴⁷ The hyperbranched polymer had a lower viscosity, lower crystallinity, and a higher degradation rate than the corresponding linear polymers. The polymer was used as a carrier of the environmentally friendly antifoulant DCOIT to achieve sustained and stable release. This system had excellent anti-biofouling ability after immersion in the East China Sea for 90 days.

When a polymer is physically mixed with antifoulants, phase separation often occurs, leading to poorly controlled release of antifoulants. We grafted high-efficiency organic antifoulant 4-bromo-2-(4-chlorophenyl)-5-trifluoromethyl-1H-pyrrole-3-carbonitrile (Econe) with butyl methacrylate and PCL- vi_2 to prepare a degradable hyperbranched polymer with anti-biofouling function via RAFT polymerization.⁴⁸ As the degree of branching increased from 0 to 3.7, the degradation rate gradually increased, and the molecular weight of the degradation product decreased from 12 000 to 2100 g/mol, indicating that the polymer can be rapidly degraded into small molecular fragments. Econe can also achieve controlled release through cleavage of the main chain ester bond, so the system has excellent anti-biofouling properties. This system provides a new route to improve the static anti-biofouling performance.

As mentioned above, poly(ester-co-acrylate) with HIZs can effectively inhibit protein adsorption and microbial adhesion, and it has low swelling and good mechanical properties. To

further improve the anti-biofouling ability, a hyperbranched copolymer was synthesized via RAFT polymerization of PCL- vi_2 , TCB, and MMA.⁴⁹ The hyperbranched structure can not only effectively reduce the crystallinity of PCL and increase the degradation rate but also generate smaller fragments after degradation, further reducing the impact on the environment. The contact angle test showed that the hydrophilicity of the coating increased after immersion in ASW. The hyperbranched polymer containing TCB side groups demonstrated increased water solubility of the fragments after the degradation and hydrolysis, raising the self-renewal capacity, that is, the self-renewing performance was slightly influenced by water flow. Thanks to the successive PCL degradation and TCB hydrolysis, the coating could continuously generate a self-renewing zwitterionic fragmented surface and effectively inhibit the adhesion of proteins, marine bacteria, and diatoms. Since the degradation and hydrolysis reactions only occur on the surface of the coating, the hydrophobic coating maintained its excellent mechanical properties (adhesion strength of ~ 2.0 MPa) during service. Note that the types of cleavable branching points are critical to optimize the self-renewing and anti-biofouling performance. Different divinyl monomers were used as branching points including methacrylic anhydride (MAAH), N,N' -adipic bis(diacetone acrylamide hydrazone) (DAA_2H), and ethylene glycol dimethacrylate (EGDMA).⁵⁰ As MAAH content increased, the degradation rate of the polymer increased, and it showed the following trend: $\text{EGDMA} < \text{MAAH} < \text{DAA}_2\text{H}$. Antibacterial and antidiatom assays indicated that the coatings efficiently inhibited the adhesion of the marine bacteria *Pseudomonas* sp. and the diatom *Navicula incerta* because of the synergistic effects of branching point cleavage and zwitterion self-generation. In short, we can control the self-renewal rate of hyperbranched polymers by tuning the content or type of branching points to meet different application requirements in marine environments.

The diversity of fouling organisms and the complexity of marine environments make anti-biofouling a challenge. High-

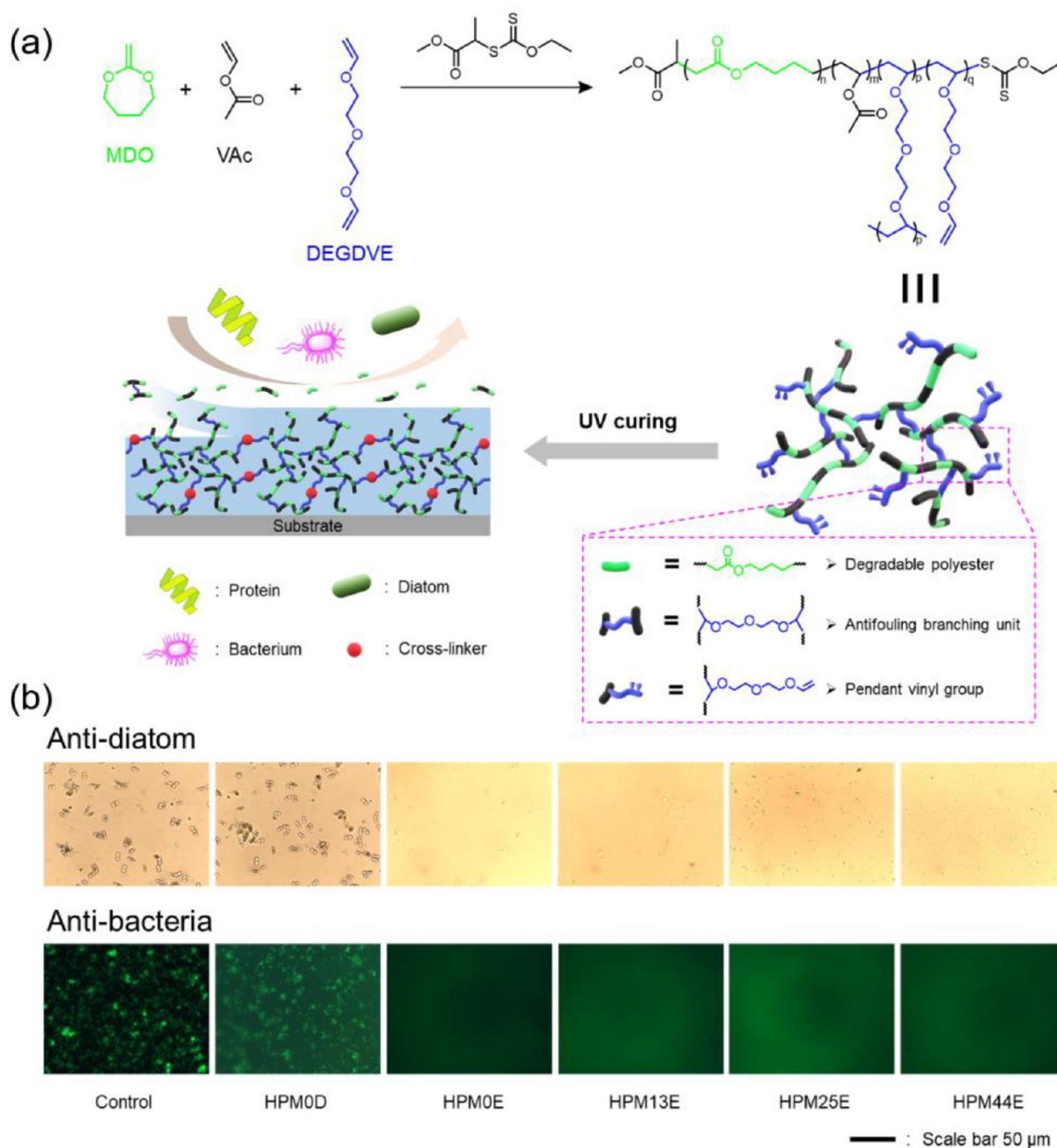


Figure 5. (a) Scheme of UV curable degradable hyperbranched poly(ester-co-vinyl). (b) Photographs of *Navicula incerta* and fluorescence microscopy photographs of *Pseudomonas* sp. adhered on the polymeric coating surfaces. Adapted with permission from ref 52. Copyright 2021 Royal Society of Chemistry.

efficiency anti-biofouling is unlikely to be realized relying on a single anti-biofouling mechanism. We proposed a new strategy referred to as the “kill–resist–renew trinity”,⁴ which integrates fouling organism killing, resisting, and releasing functions (Figure 4). A dual-functional monomer, tertiary carboxybetaine ester acrylate with anti-biofouling group *N*-(2,4,6-trichlorophenyl)maleimide (TCB-TCPM), was synthesized and copolymerized with MAAH via RAFT polymerization. Before hydrolysis of TCB-TCPM, the polymer showed excellent anti-biofouling ability because of the contact-kill mechanism from TCB groups. After hydrolysis, TCB-TCPM released the antifoulant TCBM to kill fouling organisms while generating zwitterions that effectively inhibited the adhesion of proteins

and bacteria (Figure 4a). Moreover, the MAAH branching point allowed the hyperbranched polymer to degrade into small molecular oligomers, and the mass loss increased with MAAH content, forming a continually renewing surface that can release dead bacteria and sea mud into the marine environment (Figure 4b). As shown in Figure 4c,d, such a hyperbranched polymer with multiple functions has excellent anti-biofouling ability and shows great potential in marine anti-biofouling.

4.2. Hyperbranched Poly(ester-co-vinyl) with Degradable Main Chains

Besides cleavable branching units, hyperbranched poly(ester-co-vinyl) with degradable main chains was synthesized via controlled radical polymerization of MDO and divinyl adipate.⁵¹

Recently, we have synthesized hyperbranched poly(ester-co-vinyl) with degradable main chains via radical ring-opening hybrid copolymerization of MDO and diethylene glycol divinyl ether (DEGDVE) for marine anti-biofouling (Figure 5a).⁵² The hyperbranched polymer has randomly distributed ester bonds because DEGDVE has reactivity similar to MDO, so it degrades uniformly in seawater. Thanks to the presence of unreacted vinyl groups in the hyperbranched polymer, the coating can form a cross-linked network after UV curing and has adhesion strength of 1.0 MPa. The hyperbranched polymer coating with 44 mol % ester has a mass loss of 1.0 mg·cm⁻², whereas the polymer coating without ester has a mass loss of 0.4 mg·cm⁻² after immersion in ASW for 14 days. The degradation could be accelerated with lipase. The final degraded product has a molecular weight of only 400 g/mol and is completely soluble in seawater, which promotes surface self-renewing without marine microplastics. Anti-biofouling tests demonstrated that the polymer with 50 mol % ethylene glycol units exhibits excellent antiadhesion performance in relation to proteins, marine bacteria, and diatoms (Figure 5b). That is because hydrophilic ethylene glycol forms a hydration layer on the coating surface that increases the adhesion energy required for biofouling. We further concentrated on the synthesis of functional hyperbranched poly(ester-co-vinyl) and successfully incorporate HIZs into the main chains, which can hydrolyze and switch into zwitterions, exhibiting excellent anti-biofouling ability. Being efficient and environmentally friendly, hyperbranched poly(ester-co-vinyl) with degradable main chain is expected to play an important role in marine anti-biofouling.

5. PERSPECTIVE AND CONCLUSIONS

Degradable vinyl polymers represent a new generation of anti-biofouling materials after self-polishing copolymers and degradable polymers. A series of main-chain degradable and side-group hydrolyzable polymers with excellent surface renewability, mechanical properties, and anti-biofouling performances have been prepared. They can be synthesized via hybrid copolymerization or RAFT polymerization. Particularly, we can regulate the polymer structure from linear to hyperbranched, the latter having short main chain segments and cleavable branching units or segments. Degradable vinyl polymers can also load green antifoulants with controlled release so that the service life can be prolonged. Since the main chains of the polymers degrade into low molecular weight fragments, they do not form the so-called marine microplastics.

Although notable progress has been made in the development of high-performance marine anti-biofouling polymers, we still face the following challenges: (1) increasing the versatility of hybrid copolymerization to synthesize degradable vinyl polymers with well-defined structure; (2) discovering new cleavable monomers to prepare degradable vinyl polymers with more controlled cleavable groups; (3) utilizing biosourced cyclic monomers instead of petroleum-based ones to make the degradable vinyl polymers environmentally friendly and realize the aim of carbon neutrality; (4) synthesizing new functional vinyl monomers to prepare degradable vinyl polymers with multiple functions such as intelligent response, drag reduction, anticorrosion, self-healing, self-cleaning, and so on. With the range of new-generation anti-biofouling polymers still expanding enormously, it is likely that these goals will be achieved in the near future. We believe that marine anti-biofouling with degradable vinyl polymers will bring this field into a new era. We hope that this Account will act as a trigger for additional

captivating thoughts among the readership in fields related to anti-biofouling such as wastewater treatment, biomedical materials, and so on.

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

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